Beyond Your Imagination

Innovate and Lead

Embedded IoT Systems

The Builder's Blueprint: From Edge Hardware to Cloud Architecture



Program: BSCS 5th Semester

Course Code: CSE-3079

Credit Hours: 3+0

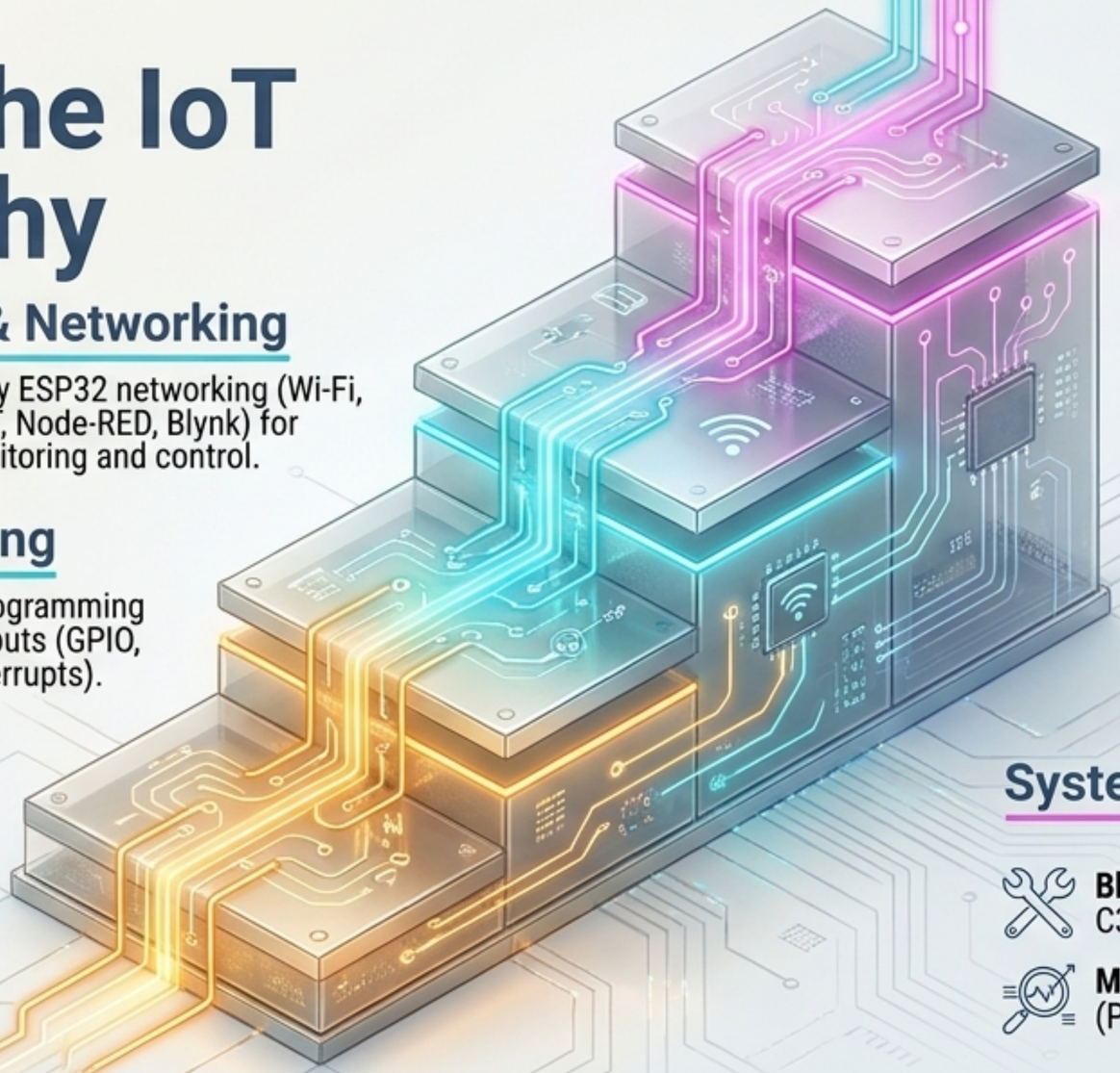
Duration: 16 Weeks

Scope: 48 Lectures (16 Tasks)

Department of Computer Science

NTU
INNOVATE AND LEAD

Ascending the IoT Skill Hierarchy



System Architecture

CLO-4: Design and implement an integrated multi-sensor IoT solution with dashboards and automated control.

Bloom Level: C6 (Create)

Mapped PLO: PLO-3 (Design/Development of Solutions)

Cloud & Networking

CLO-3: Apply ESP32 networking (Wi-Fi, HTTP, MQTT, Node-RED, Blynk) for remote monitoring and control.

Edge Interfacing

CLO-2: Apply ESP32 programming to interface inputs/outputs (GPIO, ADC, PWM, Timers, Interrupts).

Hardware & Platform Foundations

CLO-1: Explain basic concepts of IoT hardware, sensors, comms, and platforms.

Bloom Level: C2 (Understand)

Mapped PLO: PLO-1 (Computing Knowledge)

System Anchitecture

Bloom Level: C3 C3 (Apply)

Mapped PLO: PLO-2 (Problem Analysis)

Department of Computer Science

NTU
INNOVATE AND LEAD

The 4-Phase Progression Roadmap

Phase 1 (Weeks 1–4)
Embedded IoT Foundations

Core Focus
Development environments, virtual prototyping, and digital I/O.

Key Technologies
ESP32, PlatformIO, Wokwi, GPIO, Debouncing, DHT11 Sensor.

Phase Outcome
Reliable edge data acquisition and digital control.

Phase 2 (Weeks 5–8)
Embedded Sensing & Control

Core Focus
Advanced interfacing, analog data, and baseline connectivity.

Key Technologies
I²C/OLED, PWM, Timers, Interrupts, ADC, LDR, Wi-Fi, HTTP.

Phase Outcome
Autonomous local environmental response and basic web transmission.

Phase 3 (Weeks 9–13)
IoT Networking & Platforms

Core Focus
Messaging protocols, cloud logic, and remote dashboards.

Key Technologies
MQTT, Node-RED, Blynk, IoT Automation.

Phase Outcome
Bidirectional cloud-to-edge remote control and data visualization.

Phase 4 (Weeks 14–16)
Integration & Project

Core Focus
System synthesis, reliability testing, and final deployment.

Key Technologies
Multi-sensor integration, End-to-End Architecture.

Phase Outcome
A functional, production-ready capstone IoT ecosystem.

Department of Computer Science

NTU
INNOVATE AND LSAD

Assembly Schedule: Edge Hardware to Network Gateway

Week 1 | Task 1: **ESP32 Environment** (PlatformIO, LED Blink)

Week 2 | Task 2: **Virtual Prototyping** (Wokwi Simulation & Debugging)

Week 3 | Task 3: **Digital I/O** (GPIO Control, Pull-ups, Logic States)

Week 4 | Task 4: **Debouncing & Digital Sensors** (Software Debounce, DHT11 Temp/Humidity)

Week 5 | Task 5: **I²C Protocol** (SSD1306 OLED Interfacing & Sensor Display)

Week 6 | Task 6: **Event-Driven Logic** (PWM Brightness, Hardware Timers, Interrupts)

Week 7 | Task 7: **Analog Sensors** (ADC Resolution, LDR Voltage Dividers, Thresholds)

Week 8 | Task 8: **Wireless Foundations** (Wi-Fi Connections, HTTP GET Requests)

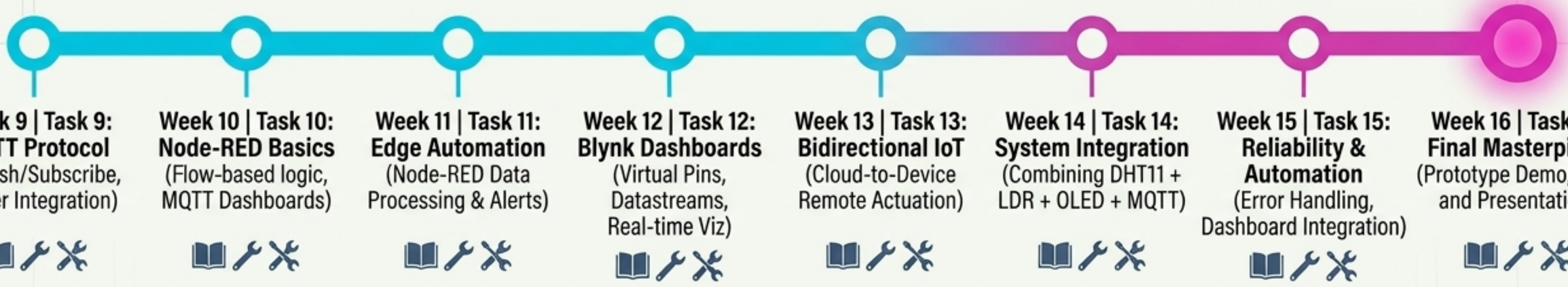
Weekly Rhythm: Every concept introduced in **Theory** (Lecture 1) is immediately built, tested, and debugged in **Practical Labs** (Lectures 2 & 3).

Department of Computer Science

subway map

NTU
INNOVATE AND LEAD

Assembly Schedule: Cloud Platforms to Capstone



Week 9 | Task 9: MQTT Protocol
(Publish/Subscribe, Broker Integration)

Week 10 | Task 10: Node-RED Basics
(Flow-based logic, MQTT Dashboards)

Week 11 | Task 11: Edge Automation
(Node-RED Data Processing & Alerts)

Week 12 | Task 12: Blynk Dashboards
(Virtual Pins, Datastreams, Real-time Viz)

Week 13 | Task 13: Bidirectional IoT
(Cloud-to-Device Remote Actuation)

Week 14 | Task 14: System Integration
(Combining DHT11 + LDR + OLED + MQTT)

Week 15 | Task 15: Reliability & Automation
(Error Handling, Dashboard Integration)

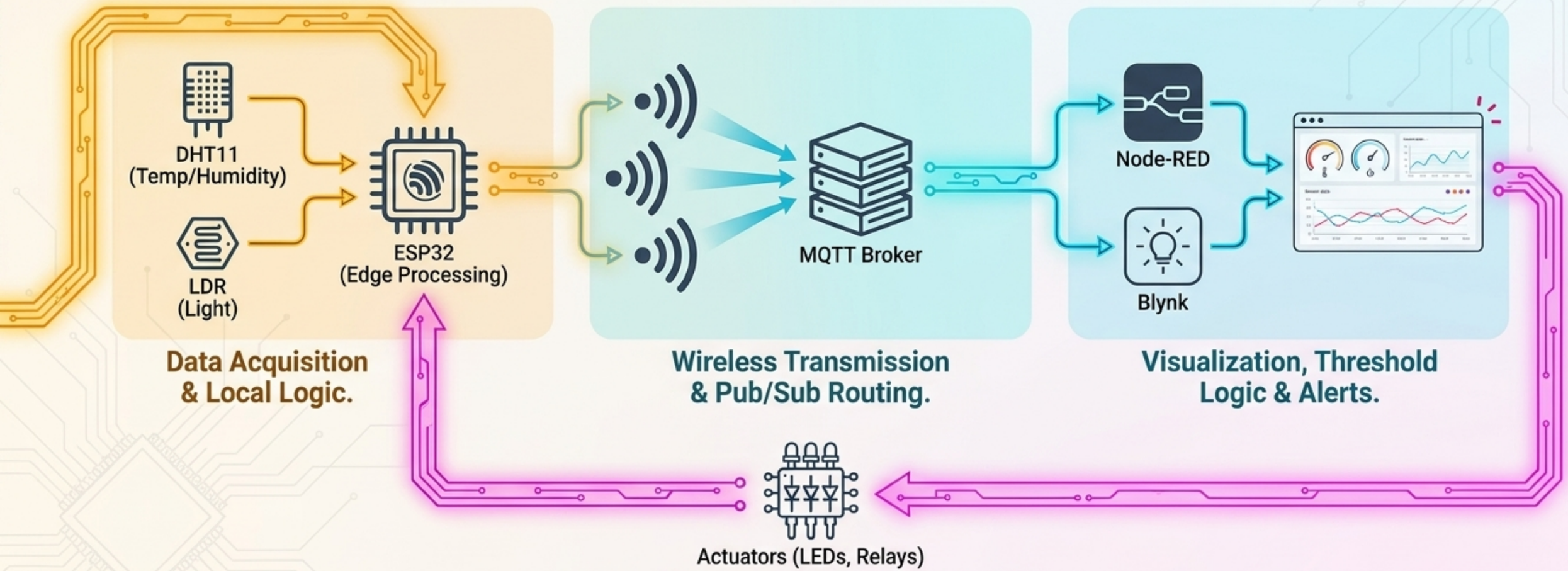
Week 16 | Task 16: Final Masterpiece
(Prototype Demo, Viva, and Presentation)

The culmination of weeks 9-16 focuses on advanced cloud connectivity, robust integration, and the final capstone project delivery.

Hardware Kit Components

Component	Qty.	Purpose
ESP32 DevKit	1	Main controller
USB Data Cable	1	Programming / power
Breadboard	1	Circuit prototyping
Jumper Wires	1 set	Connections
LEDs	2–3	GPIO / PWM output
220 Ω Resistors	3–5	LED current limiting
Pushbuttons	2	Digital input
DHT11	1	Temperature / humidity sensing
LDR	1	Analog / light sensing
10 kΩ Resistor	1–2	LDR voltage divider
0.96" SSD1306 OLED (I²C)	1	Display + I²C
Buzzer	1	Simple alert extension

The Masterpiece: End-to-End System Architecture



Department of Computer Science

The Builder's Toolkit & Resources

NTU

1

Core Texts



Core Texts

Primary: *Internet of Things: A Hands-On Approach* by Bahga & Madisetti.

Supplementary: *Pengantar IoT Praktis* by Darmawan & Packt's Node-RED guides.

2

Official Web Architecture



Official Web Architecture

ESP32 Arduino Core & PlatformIO

Wokwi (Virtual Prototyping)

Node-RED & Blynk Official Documentation

3

Video Channels



Video Channels

Espressif Systems

HiveMQ

Node-RED

Blynk IoT

Resource Use Policy

Professional Standard Required: Treat official web documentation as your primary source of truth. ESP32 APIs, PlatformIO, and Node-RED update frequently. YouTube is supplementary—always verify against current official docs.